

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.541 to 0.616
- None of 20 transferability diagnostics is shown to order transfer skill

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

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Abstract

Pre-fire thermal dryness separates a fire year from a normal one, but such models are rarely tested outside the region where they were fitted. Six thermal predictors were added to a terrain, fuel and greenness baseline in five Mediterranean regions, with MCD64A1 labels and spatially blocked validation. The first result is **evaluation geometry**: with model, predictors and fitting held fixed, scoring on the burn scar and its 2 km collar rather than region-wide costs **0.143 ROC-AUC** [**+0.077**, **+0.208**]. A control isolates the cause as the negative pool's composition, not class balance. Applied between regions the same effect withdrew five of our own claims: the study areas enclose very unequal far fields, and equalising them to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.541 to 0.616 and leaves no between-region sign reversal supported under our own criterion, though the LST anomaly still differs on a difference interval. Local skill nonetheless does not travel. The thermal block adds +0.045 to +0.148 within regions at 5 km blocking, but +0.022 [−0.032, +0.077] when a whole burn scar is withheld and +0.004 [−0.028, +0.036] across twenty transfer directions, with a sign

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that varies by pair. The static baseline transfers no better, at 0.537 against 0.541, and equalised transfer sits 0.155 below a matched within-region reference. Of twenty candidate diagnostics from five families, none was shown to order transfer, and the two that appeared to do so need target labels in both regions. Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns (Pausas and Keeley, 2021). The dominant pattern in fire susceptibility mapping (Vibhandik et al., 2026; Jodhani et al., 2026) is settled: geospatial predictors are assembled over a study region, paired with a historical burned-area record and fitted with a supervised classifier, most often a random forest (Breiman, 2001), and the surface is published with a cross-validated AUC of 0.85 to 0.95. **That pattern does not report portability**, and this paper measures it.

1.1. *Portability goes unmeasured*

A susceptibility model’s reported skill is almost always a *within-region* estimate: held-out folds come from the same study area and season, often the same fire, and random folds over cells let spatial autocorrelation inflate it further. The problem is documented across ecological modelling (Roberts et al., 2017; Ploton et al., 2020) and addressed by spatially blocked cross-validation (Valavi et al., 2019; Meyer et al., 2018), but blocking does not make the estimate honest about a fire the model has not seen.

Because only that side of the ledger is reported, portability is never entered: a predictor block is adopted on the increment it delivers inside its training footprint, and whether that survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger indices are known not to port cleanly in a Peruvian case study (Podschwit et al., 2022), and the two studies that test transfer systematically (Dimarco et al., 2026; Liu and Mostafavi, 2025) both report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific class behaves the same way is this paper’s question.

1.2. Pre-fire thermal dryness is the natural test case

Pre-fire thermal dryness is the dynamic class most plausibly *expected* to transfer. Standard predictors are static or near-static over the timescale at which fire danger varies, so they explain poorly why one summer burned and the preceding one did not; what changes is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The physics linking moisture stress to combustion is universal, so portability should be most expected here, which makes the class diagnostic: a loss cannot be dismissed as a peculiarity of a locally defined covariate. The expectation is sharpest for the internally normalised channels, which should be least exposed to absolute-temperature offsets between regions; **they transfer no better than the absolute ones** (Appendix A(f)). The expectation motivates the design and is not a finding of it — Section 4.4 reports the associations running the other way.

1.3. *Why the loss is invisible to the diagnostics in use*

The field’s instrument for anticipating transfer failure is predictor-space
45 dissimilarity — the area of applicability and its relatives (Meyer and
Pebesma, 2021, 2022; Ludwig et al., 2023) — which asks whether a target’s
predictor values fall inside the training envelope. That is a question about
the marginal distribution of the predictors, and it cannot see a change in the
relationship between predictors and response, because detecting one requires
50 the labels being withheld. If the failure is conditional rather than marginal,
those diagnostics look in the wrong place; Section 4.6 tests whether they do.

1.4. *Contributions*

Three findings carry this paper, stated here as claims and established in
Section 4, which carries every interval.

55 **Contribution 1. Where a model is scored decides what it ap-
pears to know, and the effect is large enough to dissolve findings
of our own.** That evaluation extent inflates AUC is established in species
distribution modelling (Section 2.4); what is new is a magnitude under a
controlled design. Holding model, predictors and fitting fixed and changing
60 only which cells are scored costs **0.143 ROC-AUC**, the size of the incre-
ments this design measures for a predictor block, with the cause measured as
the negative pool’s composition rather than class balance (Section 4.3). Ap-
plied between regions it withdraws five claims of our own, including the sign
reversal we had offered as the mechanism and the one arm that held place
65 fixed (Section 4.4), and the practical consequence is a reporting standard
(Section 5.7).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-region increment in all five regions, every bootstrap interval above zero, and stays positive when the frame is equalised — but much of it is a property of interleaved holdout, and across twenty transfer directions the paired contribution spans zero on both frames under the primary resampling unit, though much closer to the boundary once equalised and not under every admissible unit (Section 4.4), with a sign belonging to the pair rather than the block. Two controls bound this: the static baseline transfers no better on either frame, so the failure is not the thermal block’s peculiarity, and on matched frames and blocking equalised transfer still falls **0.155** short of the within-region reference. The within-region half is not novel (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022); the paired contrast against portability is.

Contribution 3. The shortfall cannot be anticipated by any diagnostic we could run. Of twenty candidates from five families, eighteen were not shown to order the matrix on the frames as drawn. Two conditional variants did, but both need burned labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those** (Section 4.4). The point survives without any ranking: at the point estimates the pair with the highest burned-niche overlap is among the weakest in the matrix while the lowest transfers above chance in both directions, contradicting an expectation the fire literature carries (Dimarco et al., 2026; Liu and Mostafavi, 2025) and agreeing with species distribution modelling (Vesk et al., 2021; Rousseau and Betts,

2022).

Two consequences follow, as supporting results: label-free alignment by standardisation and covariance alignment (Sun et al., 2016) does not repair
95 transfer, in what we believe is its first application to fire susceptibility; and because the residual is conditional, the resource that closes it is target labels, priced in Appendix A(u).

The leakage-audited, spatially blocked protocol is released with code, configuration and frozen outputs, so most of this can be re-run rather than
100 taken on trust, the release being complete as the declarations record — which matters given evidence that wildfire transfer conclusions are sensitive to evaluation design (Xu et al., 2026). A companion paper treats the observational layer beneath this one.

2. Related work

105 2.1. Fire susceptibility mapping, and what it reports

The dominant pattern is described in Section 1; the figure it publishes, a cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Comparable within-region results already exist for the landscapes studied here (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022).

110 2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through fuel moisture content (Yebra et al., 2013). Chuvieco et al. (2004) established the pairing of land surface temperature with a vegetation index as a live-fuel-moisture estimator for fire-danger rating, and the

115 Temperature-Vegetation Dryness Index (Sandholt et al., 2002) formalises
that feature space into an internally normalised measure, which is why
it is often expected to travel better than raw temperature. That pre-fire
thermal state carries information about subsequent fire is established (Maffei
et al., 2018; Maffei and Menenti, 2019; Maffei et al., 2021), and Gelabert
120 et al. (2025) found dead fine fuel moisture and its anomalies the most
influential predictor of human-caused ignition likelihood across Europe,
testing generalisation by pooled fitting with per-site evaluation. What
has not been tested is whether such a classifier survives strict, label-free
application to an unseen region.

125 2.3. *Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill
estimates (Roberts et al., 2017; Ploton et al., 2020), and spatially blocked de-
signs are the standard remedy (Valavi et al., 2019; Meyer et al., 2018), though
they have themselves been argued to introduce pessimistic bias (Wadoux
130 et al., 2021; Milà et al., 2022; de Bruin et al., 2022). This paper blocks
throughout and reports every result at three block sizes, so a reader can see
which verdicts depend on the choice.

Blocking corrects the inflation random folds produce within a region.
It does not speak to performance on a fire the model has not seen, which
135 Section 4.3 measures on identical cells. For that, the field’s instrument is
the area of applicability and related predictor-space dissimilarity measures
(Meyer and Pebesma, 2021, 2022; Ludwig et al., 2023), which ask whether
a target’s predictor values fall inside the training data’s envelope. Species
distribution modelling has examined transferability far more systematically

140 than fire has (Yates et al., 2018), and two studies there report that geographic and environmental similarity do not reliably predict transfer success (Vesk et al., 2021; Rousseau and Betts, 2022); transferability of satellite-derived empirical models is an active question in fire remote sensing itself (Marino et al., 2024).

145 The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes (Moreno-Torres et al., 2012). Covariate shift is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL (Sun et al., 2016),
150 and domain adaptation has an established remote-sensing literature (Tuia et al., 2016; Persello and Bruzzone, 2012). Concept shift is not: no realignment of inputs can repair a reversal in the sign of an association, because detecting one requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new (Huang et al., 2026), but we are aware
155 of no prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.4. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. (2022) report, in a Peruvian case study, that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome (Liu
160 and Mostafavi, 2025) runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label is a principal-component composite of hazard *indicators*

rather than observed burned area, and it applies no adaptation. Xu et al. (2026) argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. (2023) provide Mesogeos, a 1 km Mediterranean datacube.

Evaluation extent and AUC. Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. (2008) make it the fifth and, by their own ranking, most important reason to distrust AUC comparatively: the extent of the modelled area governs how many easy absences enter the calculation, and therefore the score. VanDerWal et al. (2009) show the same lever on the calibration side, and Barve et al. (2011) give the argument its general form as the accessible area. That literature is qualitative about magnitude, because magnitude is problem-specific, so our Contribution 1 is a measurement inside that framework rather than a new phenomenon. The wildfire literature has largely not imported the lesson: region-wide figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells — which is why Section 5.7 treats the 0.143 as a reporting problem rather than a caveat.

The nearest neighbour, and the contrast this paper draws. Di-marco et al. (2026) is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4×4

190 matrix, every transfer exceeding AUC 0.80 and bioclimatically similar coun-
tries scoring higher, with no domain adaptation.¹ Much of the design is
shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets,
tree ensembles, spatially aware validation, an explicit transfer matrix — and
two things differ. **The predictor class:** their model rests on attributes of
195 a place, all spatially stationary, ours on the state of a surface in one season.
The response variable: theirs is human-driven ignition, ours burned area,
which have different dominant controls. Read together the two results sug-
gest the relationship between domain similarity and transfer success may be
predictor-class dependent, with Vesk et al. (2021) and Rousseau and Betts
200 (2022) aligning from species distribution modelling while Dimarco et al. and
WildfireGenome run against it. We present that as a live disagreement; Sec-
tion 5.6 states why our data do not settle it.

3. Data and methods

3.1. Study regions and temporal windows

205 Five Mediterranean wildfire regions are analysed (Fig. 1): Manavgat
2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in
Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangle in
EPSG:4326, defined from place coverage rather than from a fire perimeter
and deliberately not clipped to it, so that unburned cells around each fire
210 form the negative class; none was tuned on burned prevalence, gate outcome

¹Their Results text refers to “LST anomalies”, inconsistent with their own Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology (Muñoz-Sabater et al., 2021). We follow their Methods and note the discrepancy so the comparison is transparent.

or any model metric. **One choice was label-informed and is stated as such:** the North Evia box was extended after the legacy box proved atypically high in burned prevalence, the extended geometry then defined from place anchors and the legacy variant kept as a sensitivity arm. Section 4.4 shows this framing decision is consequential and Appendix C.5(x) treats it as the design lesson of the paper. A sixth region, Kozan 2023, is carried as a negative control and excluded by the gate of Section 3.3.

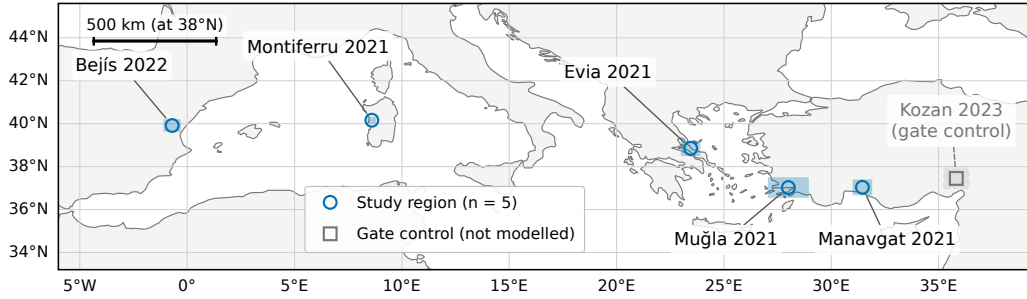


Figure 1: Study regions and the gate negative control. Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table C1 to 5×10^{-3} degrees.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Lengths vary with the event: 57 to

61 days for predictors, 35 to 59 for labels, counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Appendix C, Table C1. The processing chain and the evaluation programme it feeds are drawn in Fig. 2.

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 (Giglio et al., 2018) retrieved through Google Earth Engine (Gorelick et al., 2017), whose omission and commission characteristics (Boschetti et al., 2019) bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the label window; the label never affects eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover

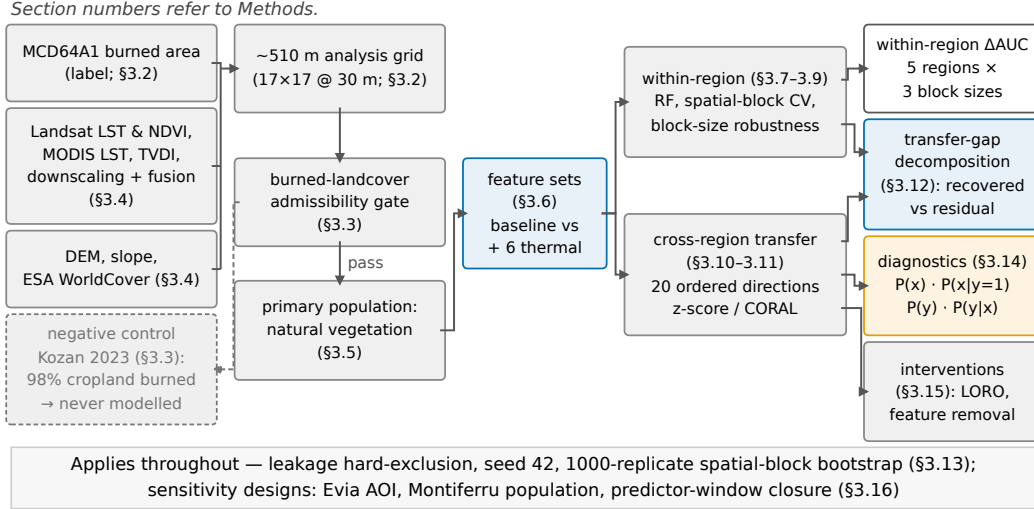


Figure 2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

classes (Zanaga et al., 2022) on the same cells. A region is admitted at 0.50
 245 with at least 30 burned cells, and rejected as a cropland-dominated control
 at 0.50 cropland. Verdicts and the purpose of the gate are in Section 4.1, the
 full rule in Appendix C.1.

3.4. Predictor variables

Two feature sets are used. The **baseline** is terrain, fuel and green-
 250 ness: elevation, slope, dominant land cover and a predictor-window median
 vegetation-index composite, all static or near-static over the timescale at
 which fire danger varies. The **thermal** set adds six pre-fire channels: current
 land surface temperature, its anomaly against a four-year window-symmetric
 baseline at the same cell, the Temperature-Vegetation Dryness Index and its
 255 difference against that baseline, and two coordinate-informed products, a
 downscaled and a fused surface temperature. The two differenced channels
 are the ones constructed to isolate the dynamic anomaly. TVDI’s wet and
 dry edges are percentiles of the values a given area and window happen to
 contain, so **it is not portable as a physical quantity independently of**
 260 **any concept shift** (Appendices C.4, C.5).

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~510 m cell from valid 30 m pixels
 only, with the valid fraction recorded; a cell is **valid_for_modeling** when
 it meets a 30 % valid-pixel floor and no thermal channel is missing. The
 265 **primary** population is natural vegetation, cells whose combined tree, shrub
 and grass fraction reaches 0.50, which excludes cropland from every burnable
 mask. The **secondary** population is all valid cells; the frozen export carries it

for the within-region arm in two regions, reported as a sensitivity in Appendix A(v), and the transfer matrix is defined on the primary population only.

270 3.6. Classifier

The two feature sets of Section 3.4 are nested, the thermal set being the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest (Breiman, 2001) with 300 trees, unlimited depth, `min_`
275 `samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison here is confounded by a model choice.

3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis
280 grid formed as `row_500m // B by col_500m // B`, and five folds are drawn over whole blocks, so no block is split between training and test. B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the
285 interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

290 For each ordered pair the model is fitted on the source population and applied to the target with **no target label used at any point**: no refitting,

no threshold selection, no calibration. Both feature sets are transferred, so the thermal block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions
295 are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics
300 from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

CORAL after region-wise z-score. The source covariance is aligned to the target’s by the standard whitening-recolouring map (Sun et al., 2016),
305 $\lambda = 10^{-5}$. Critically **the transform is applied to the source only**; the target is left as it is, and the classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time. λ sensitivity was assessed over nine values on four of the twenty directions, moving transfer AUC by at most 0.014; the canonical $\lambda = 1$ lies outside that
310 sweep (Appendix A(b)).

3.10. Transfer-gap decomposition and the concept-shift criterion

The transfer gap is decomposed as (adapted – raw) / (within – raw), signed and unclipped, with its interval from the same paired bootstrap; Section 4.3 shows the remainder should not be read as a conditional residual,
315 because much of it is incurred inside a single region (Appendices A(j), C.7).

The mechanism is diagnosed by **signed univariate association**: the raw ROC-AUC of each numeric predictor against **burned** is computed per region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction rather than weakness. A reversal is called bootstrap-supported only
320 when the two regions' point estimates fall on opposite sides of 0.5 **and each region's own interval excludes 0.5**. That is stricter than requiring the intervals to be disjoint, and a feature failing the second condition is recorded as a point reversal only.

3.11. *Transferability diagnostics versus transfer*

325 Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap that resamples unordered region pairs with both of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity,
330 climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$, measured by Schoener's D (Schoener, 1968) and Warren's I (Warren et al., 2008); regime distance $P(y)$; and conditional direction agreement $P(y|x)$. **No family-wise error control is claimed**, and the number of variants per family is reported with the result. Appendix C.2
335 gives the full specification.

3.12. *Interventions*

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and

evaluated on the held-out region, with folds blocked as before. This asks
340 whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors reversing with bootstrap support **on the frames as drawn** are **elevation_mean** and **lst_anomaly_mean**. Section 4.4 withdraws that support under an equalised frame, so the selection rule is frame-dependent and this arm is
345 reported as a measurement under the original protocol, not as a consequence of an established reversal. Both are dropped and everything refitted, within-region and across every direction, and each is dropped singly so the cost can be attributed. The two features are selected by the same reversal analysis the result is then read against, so **both quantities are post-selection**
350 **estimates** with no correction applied.

3.13. Controls on the transfer path

Four evaluations establish what the transfer arms are measuring, all using the transfer protocol of Section 3.8 unchanged: a **within-region half-split** (modelled cells cut at the median of a grid axis, both axes and directions,
355 a split discarded when either half is single-class); **leave-one-scar-out** (each burned component of at least 50 cells, dilated by a 2 km buffer, withheld from training and used as the target); a **foreign-region evaluation** of the same held-out scar areas; and the **same blocked model restricted** to those areas, which isolates the evaluation region from the training regime. Buffers
360 of 2, 5 and 10 km were run, 2 km primary. The per-split and per-scar positive counts are unequal and bear on the interpretation (Appendix A(i)).

3.14. Leakage control and reproducibility

An explicit forbidden-column set is enforced at every model fit as an assertion rather than a convention: coordinates and their normalised forms, every burn-date and label-provenance column, and the agreement fraction
365 are excluded from all feature sets. The natural-vegetation mask defines the population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates, with one qualification: the five diagnostic bootstraps behind Table 5 and Table B1 use per-measure offsets from that seed
370 rather than the seed itself, so that independent measures do not share a re-sampling draw. Two of the twenty transfer verdicts are not stable across seeds and are identified in Section 4.5. Because the transfer analysis runs in an environment separate from the upstream pipeline’s, every within-region model was refitted against the frozen upstream output: the within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} ,
375 with the tolerance that applies if the library version is not pinned in Appendix C.5(vi). Headline results are repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four classifier capacities, and where a conclusion depends on one of those choices **the dependence**
380 **is reported rather than resolved by choosing the favourable setting** (Appendices A, C.6).

3.15. Same-geography event-to-event comparison (*Muğla 2021 versus 2022*)

Muğla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical grid,
385 eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell bootstrap used elsewhere. **Season, year and**

population all differ, since the 2022 arm is defined by removing the 2021 scar (Appendices A(m), C.3).

4. Results

390 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate and the negative control fails it as intended: Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded. The separation is not marginal, the five admitted regions carrying 0.723 to 0.991, so
395 the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, **though one negative control cannot establish it**. North Evia is analysed on an extended AOI, which leaves the burned scar essentially unchanged
400 while cutting TSG prevalence from 0.676 to 0.287; the effect on transfer is in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region, the increment's bootstrap interval
405 excluding zero in all five at 1 km and at 5 km blocking (Fig. 3) — the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154, but rest on 6 to 33 positive-carrying blocks and are indicative.

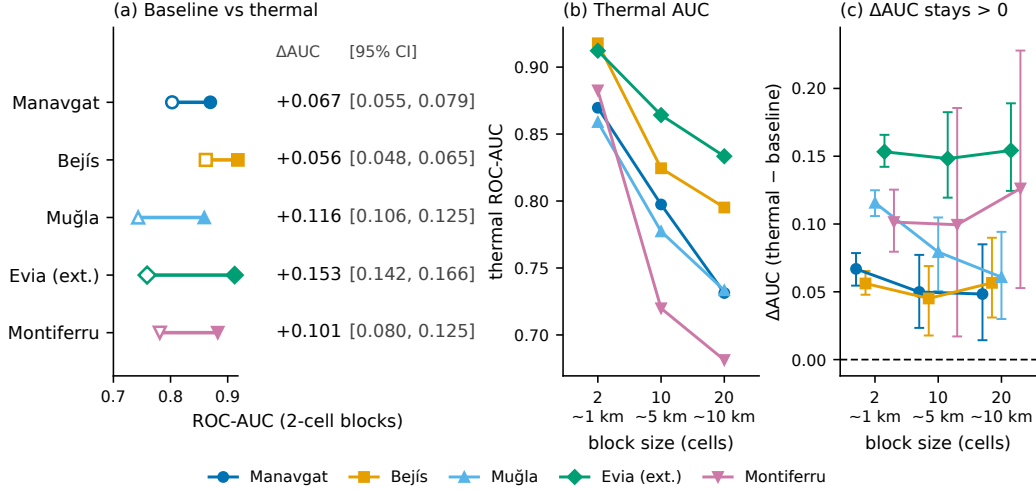


Figure 3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 1, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes (2, 10, 20 cells)**, and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
2021	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
2022	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
2021	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
2021 (ext.)	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferi	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
2021	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

Table note (resampling units). The bootstrap resamples spatial blocks,
 410 so an interval’s reliability is bounded by the number of blocks carrying at
 least one burned cell, and those counts fall from 192 to 843 at 2 cells to **6**
to 33 at 20 cells. An interval built on six such blocks has no meaningful
 coverage, so **the 20-cell row should be read as indicative rather than**
as an interval. The 10-cell row is the coarsest blocking this design supports
 415 properly, every region there having 16 to 70 positive-carrying blocks, and the
 increment holds at that scale in all five regions (Appendix A(c)).

This within-region result is not itself novel; it is reported because the
 transfer arms below are measured against it.

4.3. What the evaluation frame is worth, within one region

420 This section isolates the effect Section 4.4 applies between regions. It
 comes first because it is measured on one model in one region with no transfer
 involved, so nothing the transfer arms do can explain it (Appendix A(i)).

Where the skill is lost, on a matched comparison. Four evalu-
 ations are reported; the last three are scored on **identical cells**, so they
 425 differ only in what the model was trained on, and the first shows why an
 unmatched comparison misleads. The held-out unit is a burned connected
 component of at least 50 cells with all cells within 2 km of it. What makes
 that harder than a whole region is the composition of its negatives, not its
 burned fraction, and that is measured rather than argued. Scoring the same
 430 predictions on a random sample of region cells drawn at the scar area’s own
 burned fraction gives **0.782 against the region-wide 0.782**, so matching
 prevalence changes nothing, at $-0.000 [-0.003, +0.002]$; scoring them on the
 scar area gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. That swaps both

pools at once, so each was swapped singly: replacing only the negatives costs
 435 **0.147** [**0.104**, **0.191**], the whole of it, and replacing only the positives costs
0.002 [**-0.052**, **+0.048**], null on average though the per-scar effect runs
 from -0.12 to $+0.12$. **The effect is the negative pool** — every negative
 in a scar collar is fire-adjacent, sharing the terrain, land cover and synop-
 tic conditions of the positives, whereas a region’s negatives include its easy
 440 far field — and ROC-AUC is in any case invariant to class balance at fixed
 class-conditional distributions (`pool_decomposition.json`).

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation pop-
 ulation. Rows B, C and D are scored on the held-out scar area; row A is the whole region
 and is shown to make the mismatch visible. Means and Student t intervals are over the
 eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one-scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**, and Bejís has no
 row C because it is its region’s only component of any size, which is why
 the controls above are computed on nine and report 0.782 and 0.627 against
 445 this table’s 0.776 and 0.634. The intervals are Student t over the scars, this
 arm’s resampling unit rather than the spatial-block bootstrap used elsewhere.
 Four scars are in Muğla and two in Montiferru, and **row A is a region-**
level quantity repeated identically across a region’s scars, so its

interval is pseudo-replicated and should not be read as coverage.

450 That repetition propagates into $A - B$ and $A - C$; clustering by region gives $+0.155$ [$+0.082$, $+0.228$] and $+0.251$ [$+0.093$, $+0.409$], both still excluding zero but the second 2.3 times wider, so **the scar-level intervals on these two differences overstate precision** (Appendix A(i)).

The same model, scored two ways on the same region, differs by
455 **0.143 AUC**. Rows A and B are one model and one set of out-of-fold predictions; the only change is which cells they are scored on. That change alone costs **0.143** [$+0.077$, $+0.208$] of the 0.225 [$+0.157$, $+0.293$] fall from A to C, about two thirds — a ratio of two estimates, compatible with anything from roughly a third to nine tenths, so the point is the size of the numerator
460 rather than the precision of the fraction. It is the size of the predictor-block increments this design itself measures, so a region-wide blocked figure is an upper bound on what the same model achieves where the fire actually is.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is $+0.082$
465 [-0.011 , $+0.175$]; C minus D, the effect of replacing a same-region model with one fitted 306 to 2,802 km away, is -0.003 [-0.075 , $+0.069$]. Both span zero and both arms sit close to chance, so **with eight scars, three of them starved of positives, this design cannot establish a fire-specific residual, only bound it at about 0.18**. It is not shown to be zero on an
470 unseen fire; what this design establishes is that it is not established there. Nor does it establish the fire *event* as the unit, since the held-out patch is defined by the labels and its identity cannot be separated from its location. Appendix A(z) reports the spread behind row D and the sweeps showing the

patch definition does not drive the result.

475 **The increment declines with the holdout, and is not established
once the fire is withheld.** Contribution 1 is about the paired thermal-
minus-baseline difference, so the same evaluations were run on it: +0.056 to
+0.153 under blocked cross-validation with every interval above zero, +0.027
on a within-region half-split, **+0.022** [**-0.032**, **+0.077**] under leave-one-
480 scar-out, and +0.004 [-0.028, +0.036] across regions. The point estimate
falls monotonically as the holdout hardens and the last two intervals span
zero, so **the within-region increment is substantially a property of
interleaved holdout.**

4.4. The same effect between regions, applied to our own matrix

485 Section 4.3's effect applies with equal force between regions. This test is
reported before the transfer matrix because it changes what Section 4.5 and
Appendix A(s) can claim; it withdraws nothing in Section 4.3, which is what
it is built from. Sources are `aoi_frame_auc_frozen_mugla.csv` and `aoi_
frame_transfer_frozen_mugla.csv`, which supersede the pre-correction
490 files for the reason given below, with `collar_frame_bootstrap.csv` and
`diagnostics_collar_frame.csv`; code is under `paper/code/` and the
elaboration in Appendix A(w).

The five areas of interest are not comparable frames. Each is a
rectangle drawn around a fire, and they differ by an order of magnitude in
495 how much unburnt far field they enclose: the share of modelled cells beyond
10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejis
(Appendix B, Table B6). That far field is not neutral — in Manavgat the
median elevation of modelled cells rises from 472 m within 5 km of the fire

to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

500 **Under an equalised frame the sign reversals do not survive.** Restricting every region to cells within 10 km of any burned cell removes only far-field negatives — every burned cell is at distance zero and is retained at any radius — so the protection against a flattering radius is the sweep, not the retention of positives. On that frame all five regions agree in sign
505 on elevation, LST and TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Under this paper’s own criterion from Section 3.10, **no between-region reversal remains bootstrap-supported:** the two features straddling 0.5 at the point estimate are the internally differenced channels, and in each at least one region’s interval covers 0.5 — Evia’s
510 by 0.003. The weaker difference instrument that Table B3’s note commits this paper to does support four opposite-sided pairs on `lst_anomaly_mean`, with three caveats that keep it weak. **The honest statement is that no between-region reversal meets the strict criterion under the collar, and that the LST anomaly differs between regions on the weaker**
515 **one.** Two scope limits belong with it, both in Appendix A(l) and A(w): the interval criterion was evaluated at the 10 km collar only, though at the point estimate the two radii agree exactly; and the collar reduces two regions’ cell counts substantially, so part of the loss of support is a loss of power, against which features supported in both regions *rise* from 1.20 to 3.40 per direction.

520 **The sign they now agree on is not the one the dryness framing predicts:** at the point estimates a hotter pre-fire surface burned *less* in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute channels behave here as static land-surface descriptors

rather than as dryness. Within distance to the nearest burned cell the sign
525 crosses 0.5 in Manavgat and is attenuated to near-null in Montiferru, so in
those two it is largely the residual spatial gradient the collar was introduced
to remove. The stratifications, the correlations and the reciprocal adjustment
are in Appendix A(k).

The same test determines what the diagnostics of Section 4.6
530 **can establish.** Both candidates there with intervals excluding zero measure
agreement in the sign of each predictor’s association between source and tar-
get — built from exactly the signed AUCs shown above to be frame artefacts,
and correlated against transfer on the same unequal frames. Recomputing
both sides under the collar:

Table 3: The diagnostics that ordered transfer, recomputed on an equalised frame. Spear-
man ρ against target ROC-AUC over the ordered directions in which each is defined. The
first two rows are the two variants that cleared zero in Table B1; the third is the all-feature
cosine, which did not and is shown for contrast. Both sides are recomputed here under
one bootstrap setting, 1000 replicates, seed 42. Source `diagnostics_collar_frame.csv`
and `collar_increment_and_cosine.csv`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction, supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$)	1.0 in all 18 directions, variance exactly 0 — degenerate
Cosine, supported features	$\rho = +0.81$ ($p = 0.0005$, $n = 14$)	$\rho = -0.06$ ($p = 0.82$, $n = 18$), variance 0.00014
Cosine, all nine features	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

535 The two fail differently and both fail: the agreement fraction has no
variance left once every pair agrees, and the supported cosine keeps a trace
of variance but stops tracking transfer. The disagreements they were reading

were the far fields. **Only these two were recomputed, so the other eighteen are unknown against the equalised transfer vector rather**
540 **than shown to be null** (Appendix A(o)). This settles what Contribution 3 can claim: **no diagnostic tested here has been shown to order transfer**, and the two that appeared to were reading how the rectangles were drawn.

Two further arms move with the frame. The same-geography arm
545 is the most extreme case in the cohort — a fixed study area is not a fixed evaluation frame, and its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021 — so under the collar its two arms fall on the same side of 0.5 on elevation, the feature whose reversal we had reported. The claim is scoped: that arm is reconstructed from the 2021
550 predictors, valid for the year-invariant channels and **not** the seasonal ones, so **only elevation and slope carry a verdict, on both the reversal is removed, and on the thermal channels the arm is silent**; it rests on eleven positive-carrying blocks against a floor of sixteen (Appendices A(m), A(o)). The within-region increment, by contrast, **survives**, positive in all
555 five regions at a mean of +0.077 against +0.086 as drawn.

The transfer matrix moves as well. Restricting source and target to the same collar:

A data-provenance defect in this arm was found and corrected:
one region’s predictor file at the canonical path had come to differ from the
560 one the frozen tables were computed on, so every arm here was re-run against the frozen export. It moves forty of a hundred per-direction values by up to 0.022 and leaves **every headline quantity above unchanged to within**

Table 4: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use a 10-cell (≈ 5 km) spatial-block bootstrap on the target, 1000 replicates, seed 42; **Table B9 reports the same matrix under 2-cell (≈ 1 km) blocking**, which is why its supported counts are the larger 12 and 6 (Section 4.5). Per-direction bounds are in `aoi_frame_transfer_frozen_mugla`.

CSV.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / below	Paired thermal delta
full	full (Table B9)	0.541	14 of 20	6	9 / 4	+0.004
full	10 km	0.576	17 of 20	3	11 / 1	+0.003
10 km	full	0.570	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.616	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	19 of 20	1	12 / 0	+0.014

0.0012, with the signed AUCs of that region’s two channels moving by at most 0.008 and no verdict changing (Appendix A(w)). The table reports the corrected values, and reproduces the frozen matrix at 0.541 against Table B9’s 0.541 and 14 of 20 exactly. **The baseline control must be restated on this frame**: the static baseline transfers at 0.593 against 0.616, a paired difference of +0.023 rather than +0.004, so the control holds in kind but the gap is about six times larger once frames are comparable, and about four times at the 5 km collar.

What this settles, and what it leaves standing. Five quantities reported below are properties of the frames rather than of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of elevation, LST and TVDI as a mechanism; the sign-agreement diagnostic; the same-geography arm; and the paired thermal contribution, +0.004 as drawn against +0.023 equalised. That last quantity carries the portability null, so it is given an interval on the frame this section argues for: **+0.023 [−0.004, +0.048]** under the pair-cluster resampling of Appendix A(o). It still spans zero, so the null survives the correction — but only just, and under the al-

ternative admissible unit, clustering by target region, it does not (Appendix A(w)). What survives is the central negative result, and its size must be stated on a matched comparison: setting 0.616 against a within-region reference of about 0.87 would compare a collar number with a full-rectangle one
585 at 1 km blocking. Recomputed on the same frame at 5 km blocking that reference is 0.772, so the shortfall is **+0.155** [**+0.094**, **+0.217**], a Student t interval over the five target regions rather than the spatial-block bootstrap used elsewhere, and numerically almost identical to Section 4.3’s negative-pool effect over nine scars, which is a different quantity on a different unit.
590 It is real, and 0.155 rather than the 0.25 an unmatched comparison implies. Appendix C.5(x) records the frame as a limitation of this cohort rather than of the method.

4.5. Cross-region transfer, and what label-free adaptation does to it

Everything in this section is computed on the frames as drawn
595 **and should be read against Section 4.4**, which has shown the frames are not comparable and that the within-region reference used here is itself frame-dependent. The as-drawn matrix is reported because it is what the pre-registered protocol yields; per-direction values are in Appendix B, Table B9, and the supporting arms in Appendix A(x).

Raw transfer is heterogeneous and includes anti-predictive directions. Target AUC spans 0.326 to 0.686, and Fig. 4 gives the matrix direction by direction. The support counts depend on the blocking scale, so both are reported: at 2-cell (≈ 1 km) blocking twelve of twenty directions are above chance with interval support and six below; at the 10-cell (≈ 5
605 km) blocking this design defends, nine are above and four below with seven

uncertain, the counts Table 4 uses. Of the below-chance directions only Man-
 avgat to Bejís survives frame equalisation, and two verdicts are not stable
 across bootstrap seeds (`transfer_ci_blocksize.csv`). Even the best raw
 transfer sits far below that target’s own within-region 0.870, the raw deficit
 610 running 0.184 to 0.592 against a reference Section 4.3 shows is not matched
 to a transfer evaluation.

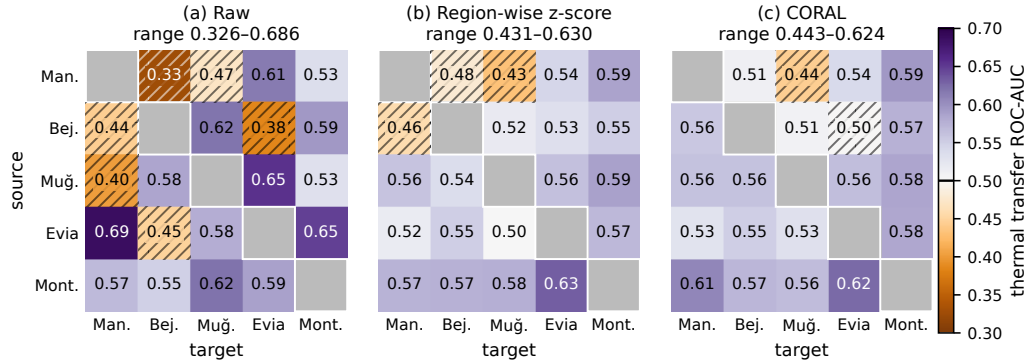


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table B9.

In precision terms it is worse than the ROC figures suggest. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. PR-AUC averages **0.156 against a no-skill baseline**
615 **of 0.136**, and **six of twenty directions fall below their own baseline at the point estimate**, five of them with intervals entirely below it; the exception is Bejis to Manavgat, whose interval covers its baseline. Only one direction exceeds twice its baseline. These are frame-as-drawn quantities and the PR arm was not recomputed on the collar (Appendix B, Table B4).

620 **The static baseline does not transfer either**, at a mean of **0.537** against **0.541** for the thermal model, so the failure is not specific to the dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel. Section 4.4 restates this control on the equalised frame, where it is 0.593 against 0.616.

625 **The thermal block’s paired contribution is +0.004, and the mean is not the informative statistic.** The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect.
630 The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from $[-0.027, +0.034]$ to $[-0.037, +0.046]$, and none propagates within-direction sampling variability. The leave-one-region-out jackknife shows how little the mean is anchored:
635 **dropping Evia alone reverses its sign.**

Label-blind adaptation compresses the matrix toward chance

rather than repairing it (Fig. 5). Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, and fourteen of twenty move closer to chance — which helps the directions that failed and harms those that worked. The committed-
640 in-advance CORAL arm averages **0.552**; taking whichever method scores better per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is **an oracle upper bound rather than an achievable result**. Even the oracle only reaches the reference a model can
645 reach on an unseen scar (Section 4.3), so alignment is regressing the matrix onto that reference rather than failing beneath it. What it cannot do is exceed the reference, and a sign reversal is not a distribution mismatch that realigning inputs would repair. Appendix A(x) reports the recovery fractions, including the seven directions with *negative* recovery.

650 4.6. Transferability diagnostics: what appears to order transfer, and why it does not

Twenty candidate diagnostics from five families were each rank-correlated with raw thermal transfer over the twenty ordered directions, under one common pair-based bootstrap.

655 **Only two diagnostics have intervals excluding zero, and both are conditional:** the sign-agreement fraction over interval-supported features at $\rho = +0.84$ [$+0.58, +0.88$], and its cosine variant at $+0.81$. No marginal measure was shown to order the matrix, including area-of-applicability dissimilarity, climatic distance and geographic distance, and neither were the

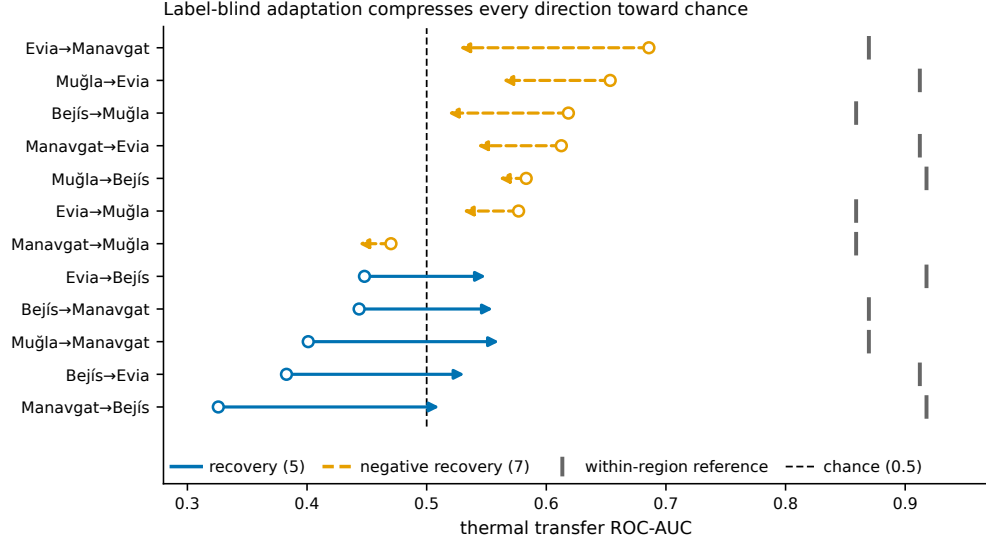


Figure 5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves away from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

Table 5: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) conditional	6	+	+0.84 [+0.58, +0.88] agreement fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain-classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] log effective-N distance	no	no
geographic	1	−	−0.24 [−0.84, +0.73] centroid geodesic distance	yes	no

660 niche-overlap and regime families. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair, so it always succeeds and carries no ordering information. Geographic separation is a single diagnostic here, centroid distance over the twelve directions for which the pipeline author’s export defines it, at $\rho = -0.24 [-0.84, +0.73]$: in the
665 expected direction and spanning zero.

Four limits are stated with the result. **Size:** the index’s tie structure caps the achievable Spearman at $+0.861$, so the observed $+0.840$ sits on that ceiling, and its exact one-sided permutation p of $0.0060 = 240$ of the $8! = 40,320$ relabellings of the **eight** pairs on which the index is defined —
670 the smallest that structure can produce, against a Bonferroni threshold of $0.05/19 = 0.0026$. **No outcome of this diagnostic could have cleared family-wise correction.** **Labels:** signed associations need burned labels in both regions, so the family that appears to work is not available before deployment while the family that fails is. **Selection,** the sharpest: the two
675 rows that clear zero are the *supported-feature* variants, whose subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint — a data-dependent selection on the same data, with no correction; their unselected counterparts over all nine features are $\rho = +0.50 [-0.17, +0.83]$ and $+0.18 [-0.40, +0.72]$, both spanning zero, so **the result lives in the selection**
680 **step.** **Frame,** established in Section 4.4, removes it entirely. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5} and leaves the ordering unchanged, so it is not an artefact of unequal samples (Appendix B). Everything here is what the original protocol yields; the conclusion that survives is that **no diagnostic tested here was**

685 shown to order transfer once the frames are comparable.

4.7. Further arms

Six further arms bear on the findings above without changing them: the contrast pair (A(s)), the two interventions (A(n)), the sensitivity summary (A(v)), the same-geography two-event arm (A(m)), the distance curve (A(t))
690 and the label-budget curve (A(u)). Each is stated there with its own limits. Three are plotted here because the shape of the result is the argument: pooling every other region never beats the best single source for any target (Fig. 8); removing the direction-reversing features costs within-region skill and returns nothing measurable on transfer (Fig. 7); and the contrast
695 pair shows the most burned-niche overlap sitting with the weakest transfer (Fig. 6). The two bearing directly on Sections 5.4 and 5.7 are the interventions, where a local cost of -0.081 is measured against a transfer return of $+0.014$ $[-0.017, +0.045]$ whose interval spans zero, and the label budget, where thirty-two labelled 5 km blocks recover 85 to 89 % of the target's
700 matched ceiling in three of six directions and 30 to 57 % in the rest — 7 to 20 % of the target's population, drawn from the event being predicted, which is not a resource available before that event burns.

5. Discussion

5.1. Reading the three findings together

705 Section 1.4 states the three findings and Section 4 establishes them; this section argues from them. One relation between them makes the paper co-

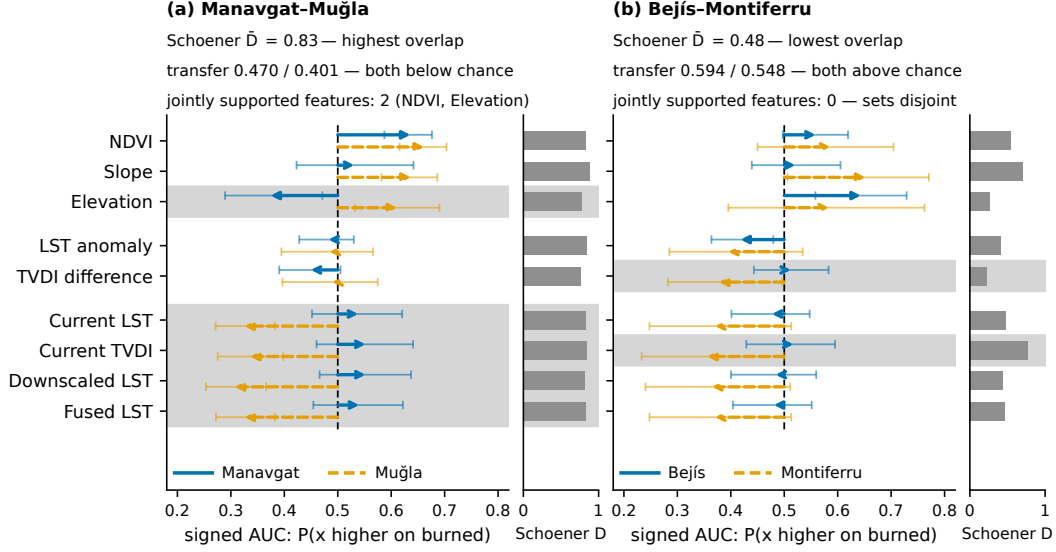


Figure 6: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region’s 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener’s D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions on the frames as drawn (0.470 and 0.401); under the equalised frame of Section 4.4 it is 0.551 and 0.510, above chance but still among the weakest in the matrix. **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 as drawn, 0.669 equalised, and 0.548 against 0.624). High niche overlap is therefore not sufficient for transfer, which is an ordinal claim on either frame. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions’ supported sets are disjoint and Montiferru’s intervals are wide. Pair (b) therefore drops out of the supported-index sample entirely. The index is undefined exactly on the pair that most challenges the overlap explanation, which is a limitation of the index and not evidence for it.

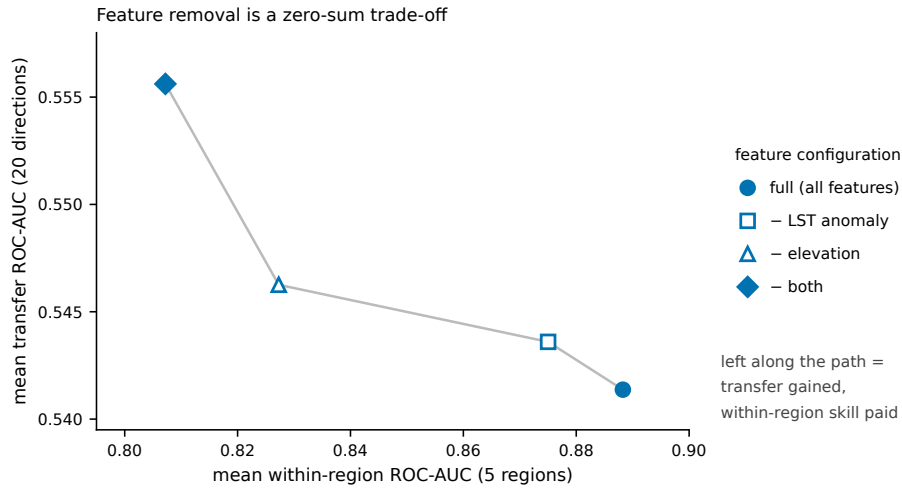


Figure 7: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

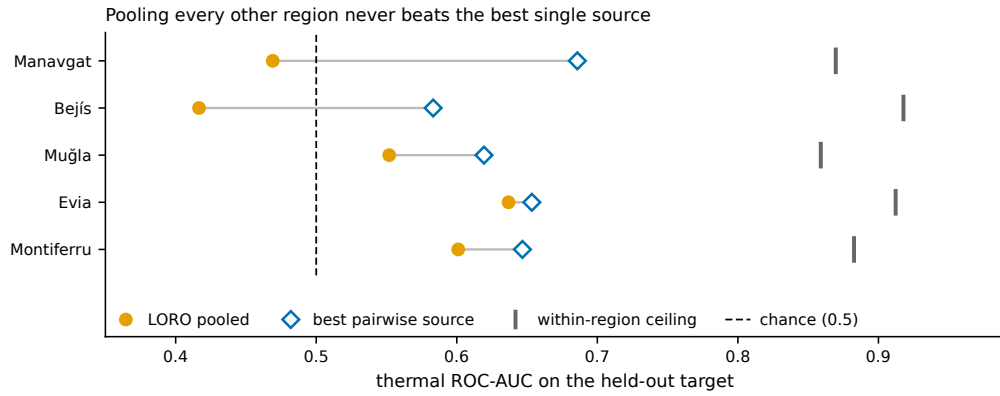


Figure 8: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

here: the first is not a caveat attached to the other two, it is the instrument that sets their size. Applied to our own matrix it withdrew five claims, and what it left standing is a shortfall in transferred skill, not a reversed
710 relationship.

5.2. *Why the thermal increment is real but local*

The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference is already present inside a single region. The increment is local in a specific sense: it
715 holds where held-out cells are interleaved with training cells and most of it is gone once they are not, before the fire or the region changes. It is not an artefact to be explained away, surviving every sensitivity arm of Appendix A(a)–(h), but it is established under interleaved validation and not beyond it — and even there the window-closure arm **weakens monotonically in**
720 **Evia**, so what holds everywhere is survival, not improvement.

The natural objection is that Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not**
725 **answer it**: Section 4.4 shows it is the most extreme frame artefact in the cohort. Three further confounds were never resolved in any case: season and year, which **cannot be resolved in this study area**, an unfixed population, and a positive-block count below this design’s own floor (Appendix C.5(ii)). With one fire per region everywhere else and that arm withdrawn,
730 **this cohort provides no evidence that the transfer shortfall is regional rather than event-specific**. Appendix A(t) adds a length scale for

the within-region decay but cannot turn it into an attribution either.

One conclusion survives from the other direction: the static baseline transfers no better than the dynamic block, so whatever the shortfall is,
735 it is not the thermal block’s peculiarity.

5.3. *Why the diagnostics fail, and what the conditional one is not*

A diagnostic built on distance in predictor space asks whether the target’s predictor values look like the training data’s, which is not the question that matters when the failure is conditional: a target region can sit well inside
740 the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the full-frame area of applicability and transfer among the weakest in the matrix **on either frame**, while the least similar pair transfers better (Appendix A(s)). High overlap does not buy transfer.
745 **The qualifier is not decorative:** at the 5 km blocking this paper otherwise defends neither contrasted direction carries a verdict, so this is a statement about point estimates. The domain classifier is at ceiling for every pair, so separability carries no ordering information either.

The two conditional variants that did clear zero are not a remedy. Section 4.6 states four limits with them: a tie-structure ceiling and a family-wise
750 threshold **no outcome could have cleared**, a label requirement that makes it a mechanism diagnosis rather than a screen, a feature subset selected on the same data, and the frame, which removes the result altogether — the index is built from signed associations Section 4.4 shows to be frame artefacts.
755 So the practical conclusion is not “use this index instead”, nor even “it works but needs labels”: **none of the twenty candidates was shown to order**

transfer, and the two that appeared to were reading how the rectangles were drawn — nulls on ten effective pairs, not shown to order transfer rather than shown incapable of it.

760 5.4. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays well below the within-region ceiling (Appendix A(n)), so it does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-
765 region skill with interval support in every region, and returns $+0.014$ $[-0.017, +0.045]$ in mean transfer.

That pair of numbers is easy to read as an exchange, and it is not one. Both arms are null on the portability axis — the thermal block’s own contribution is $+0.004$, removal returns $+0.014$, both intervals spanning zero —
770 so they measure a local cost and no compensating gain. That is not a conservation law and not a rate at which local skill can be sold for portability; no such rate is estimated here.

5.5. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null: the regime-distance correlation has the wrong sign at the
775 point estimate, the most regime-similar pair fails in both directions, and the most regime-different pair transfers above chance. One mundane explanation can be set aside: subsampling Muğla, much the largest population, to Manavgat’s cell count and separately to its positive count leaves its transfer
780 behaviour inside the subsampling range in both roles. The error was in the

hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology (Archibald et al., 2013) in general.

5.6. *The empirical contrast with Dimarco et al.*

Dimarco et al. (2026) transfer successfully across a comparable Mediter-
785 ranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours the state of a surface in one season, which suggests the relation between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here:
790 their response variable is human-driven **ignition**, dominated by access and activity, while ours is burned **area**, dominated by spread, so a predictor-class explanation and a response-variable explanation are confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this
795 cohort the place-attribute class does not travel either.

5.7. *Implications*

In precision-recall terms, which is how a susceptibility surface is used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136 **on the frames as drawn**, six of twenty falling below their own
800 baseline (Section 4.5). Whatever the ROC figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

The paper supports one concrete change in reporting: alongside a spatially blocked within-region figure, report skill on a held-out burn scar and

its surroundings. On these five regions the two differ by about 0.14 ROC-
805 AUC on the same model, the size of the effect such papers usually claim, so **a
blocked figure alone should be read as an upper bound**. Transfer skill
likewise has to be *measured* rather than inferred from similarity, since the
diagnostics used for that inference did not order it here, and where a model
must be moved the resource that closes the gap is target labels (Appendix
810 A(u)): a real answer, but not a cheap one.

The results also bound what unsupervised alignment can be asked to do.
Even the oracle selection, which uses the target labels the protocol forbids,
only reaches the reference a model can reach on an unseen scar (Section 4.5).
Alignment is not failing far below an achievable target; it is regressing the
815 matrix onto it, at the cost of the directions that already worked. A sign
reversal is not a distribution mismatch that realigning inputs repairs.

5.8. Limitations

Eleven limitations are stated in full in Appendix C.5; four bind the conclu-
sions above. **The frames are not comparable and this cohort cannot**
820 **fully repair it** (Section 4.4): the collar radius is itself a choice — 5 km and
10 km give 0.608 against 0.616 — and the frames were fixed upstream, so
we can restrict them but not extend them. Any future cohort should fix the
frame by an explicit accessible-area rule (Barve et al., 2011) before any pre-
dictor is computed; that is the main design lesson of this paper. **Each region**
825 **contributes one fire season**, so regional concept shift is confounded with
event meteorology and the shortfall cannot be attributed to region rather
than event (Section 5.2). **The diagnostic correlations rest on an effec-
tive sample of ten region pairs**, so Section 4.6’s successes and failures

both read at that power. And **the interval-support counts are less sta-**
830 **ble than the point estimates behind them:** the published split turns
on a lower bound of -0.00045 . The remaining seven, from the absent mete-
orological covariates to the single classifier family (Appendix A(h)), bear on
scope rather than on the conclusions above.

6. Conclusions

835 Where a fire model is scored decides what it appears to know. Holding
model, predictors and fitting fixed and changing only which cells are scored,
moving from a whole study region to the burn scar and its 2 km collar
costs **0.143 ROC-AUC**, with a control measuring the cause as the negative
pool's composition rather than class balance. That is the size of the predictor-
840 block increments this design itself measures, so a region-wide validation figure
should be read as an upper bound on what a model achieves where fire
actually occurs, and reported as one.

That correction is not only other people's problem. Applied to our own
five-region matrix it withdraws five of our claims. The study areas enclose
845 very unequal far fields, and equalising them to a 10 km collar lifts mean
transfer from 0.541 to 0.616, reduces the directions below chance, and leaves
no sign reversal supported between regions under our own criterion. The
same correction applies to the one arm that held place fixed, two fires in one
study area eleven months apart, which we had wrongly exempted because
850 its geography was constant while its evaluation frame was not: its elevation
reversal does not survive equalisation either, and that arm can speak only
for the year-invariant channels. What had looked like a reversed predictor-

burning relationship was mostly a statement about how five rectangles were drawn.

855 What survives is the central result. Pre-fire thermal dryness adds a real and repeatable within-region increment in all five regions, surviving spatial blocking at about 5 km and a predictor window closed up to two weeks before the first labelled burning. It does not travel: on matched frames and blocking, equalised transfer of 0.616 falls 0.155 short of the within-region reference; 860 the paired cross-region contribution spans zero on both frames under the resampling unit we treat as primary, though not under every admissible one (Section 4.4), with a sign that varies by pair; and the static baseline transfers no better on either frame, so this is not a peculiarity of thermal predictors. The shortfall also cannot be anticipated: none of twenty diagnostics from 865 five families was shown to order the matrix, and the two that appeared to require target labels on both sides and stop ordering transfer once the frames are comparable. Label-free alignment compresses most directions towards chance rather than repairing them, and supervised recalibration works but is not cheap.

870 Two limits belong with all of this. The agreed thermal direction is the opposite of the one the dryness framing predicts — hotter pre-fire surfaces burned less in every region, and holding greenness does not remove it while holding temperature reverses greenness in two regions — so on this cohort the absolute channels behave as static land-surface descriptors rather than 875 as a dryness index. And with one fire per region, and the single arm that held place fixed now withdrawn, this design measures a transfer shortfall but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may
880 contribute nothing distinguishable from zero across regions. And fix the evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both a transfer failure and a mechanism to explain it. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically
885 normalised dryness variables.

Appendix A. Appendix A. Where the rest of the appendices are

Two appendices are printed here: **Appendix B**, the diagnostic tables below, and **Appendix C.5**, the eleven limitations. They carry the per-direction numbers and the limitations against which the claims are checked,
890 so the paper can be assessed without leaving it.

Appendix A and the protocol sections C.1 to C.4, C.6 and C.7 are released with the paper rather than printed in it, at `paper/supplementary_appendices.md` in the repository named in the declarations. They keep their section names there, so a pointer such as Appendix A(c) or
895 Appendix C.2 in the text resolves in that document unchanged. Nothing in them is evidence a claim depends on that appears nowhere else; every number in them also sits in a frozen artefact named in the text. The lettering of this appendix is kept for that reason: renaming would have broken 148 references between the paper, the supplementary appendices and the supplementary
900 material.

Appendix B. Appendix B. All twenty transferability diagnostics

Section 4.6 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to
905 check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru.
910 The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n	dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16		+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16		+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20		+0.50 [−0.17, +0.83]	no
Vector	P(y x)	+	20		+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional					
Agreement count, all 9	P(y x) conditional	+	20		+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20		+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20		+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20		+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20		−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20		−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20		−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12		−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	51	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	P(x) marginal	+	12		+0.22 [−0.48, +0.59]	no
Fraction inside unweighted	P(x) marginal	+	12		+0.08 [−0.89, +0.63]	no

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it.

915 With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

920 **Reading the two rows that clear zero.** Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and

925 Section 4.6 says so.

Equal-sample check. The families sit on unequal samples: marginal, applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values

930 to 4.8×10^{-5} and leaves the ordering unchanged — the conditional rows still lead at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

Appendix B.1. B2. The signed univariate associations the reversal claim rests on

935 Appendix A(n) drops two predictors because their signed association with burning reverses between regions with bootstrap support on the frames as drawn. Section 4.4 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1,
940 not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against burned, by region. Primary natural-vegetation population; 10-cell (~ 5
945 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_	0.374 [0.289,	0.643 [0.558,	0.611 [0.532,	0.541 [0.448,	0.584 [0.395,
mean	0.471]	0.729]	0.690]	0.626]	0.762]
slope_mean	0.531 [0.423,	0.521 [0.439,	0.637 [0.582,	0.487 [0.418,	0.652 [0.506,
	0.642]	0.605]	0.686]	0.554]	0.771]
ndvi_mean	0.636 [0.587,	0.559 [0.497,	0.662 [0.616,	0.639 [0.575,	0.586 [0.450,
	0.676]	0.619]	0.704]	0.701]	0.704]
lst_anomaly_	0.482 [0.428,	0.418 [0.364,	0.485 [0.395,	0.640 [0.567,	0.395 [0.285,
mean	0.530]	0.480]	0.566]	0.710]	0.535]
current_lst_	0.538 [0.452,	0.477 [0.401,	0.325 [0.271,	0.377 [0.301,	0.370 [0.248,
mean	0.621]	0.547]	0.382]	0.456]	0.513]
current_	0.552 [0.460,	0.517 [0.429,	0.336 [0.275,	0.362 [0.285,	0.356 [0.233,
tvdi_mean	0.641]	0.595]	0.398]	0.442]	0.499]
tvdi_	0.449 [0.391,	0.512 [0.443,	0.490 [0.396,	0.519 [0.444,	0.378 [0.282,
difference_	0.505]	0.583]	0.575]	0.589]	0.497]
mean					
downscaled_	0.552 [0.466,	0.484 [0.400,	0.307 [0.253,	0.377 [0.297,	0.365 [0.240,
lst_mean	0.637]	0.560]	0.366]	0.459]	0.511]
fused_lst_	0.540 [0.454,	0.481 [0.404,	0.325 [0.272,	0.376 [0.300,	0.370 [0.248,
mean	0.622]	0.551]	0.383]	0.456]	0.513]

950 **What counts as a reversal.** A pair of regions is called a reversal

only when their signed associations point to opposite sides of 0.5 **and each region’s own interval excludes 0.5**. That is stricter than requiring the two regions’ intervals to be disjoint, and the difference matters: for `current_1st_mean` between Manavgat and Muğla the two intervals are disjoint, but
955 Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than
965 manufacturing them: a difference interval on the pair, which is the instrument Appendix A(m) uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked
970 area budget, so precision-recall is the operational quantity, and at target prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table B9.

Table B4. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided
975 by that baseline; a lift of 1 is no better than random ranking. Ordered by lift.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

Six directions fall below their own no-skill baseline, and only one exceeds
 twice it. The six are the same six that are below chance on ROC-AUC,
 which is what a reversed ranking predicts in either metric. Section 4.4 shows
 980 that this count is largely a property of the evaluation frames rather than of

a reversed predictor-burning relationship.

Appendix B.3. B5. The same-geography event pair, in full

Appendix A(m) reports this arm and Section 4.4 withdraws its **eleva-**
985 **tion** reversal as a frame artefact; Appendix A(o) explains why the thermal
channels of this arm carry no verdict either way. The per-feature values are
kept here because the arm is what motivated the frame test, and because its
structural asymmetries have no analogue in the twenty-direction matrix.

Table B5. Signed univariate feature-burned AUC, Muğla 2021
990 **versus 2022.** Raw AUC against burned, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42
(Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and
38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70**
for the 2021 arm and 11 for the 2022 arm. Table 1’s note sets sixteen
995 as the floor this design supports at that blocking, so the 2022 intervals here
fall below the paper’s own standard and are read as indicative, exactly
as the 20-cell row of Table 1 is. The 2022 arm is additionally a single
compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_`
`raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_`
1000 `reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

Appendix B.4. B6. The evaluation frames, and the signed associations they produce

Section 4.4 states these results; the per-region values are here.

1005 **Table B6. Evaluation-frame geometry of the five study regions.**
Primary natural-vegetation population. Distance is Euclidean to the nearest
burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.`
`csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

1010 **Table B7. Signed univariate AUC, frame as drawn against a 10**
km collar. Point estimates; the intervals that decide the reversal question
are given in the text below and in `collar_frame_bootstrap.csv` (10-cell
blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 -$
 $\text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no
1015 burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

Table B8. Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural-veg fraction	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Montiferru 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

1025

Table B9. Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

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Table B10. The most and least environmentally similar pairs, on both frames. Schoener’s D is computed over burned cells only and

is therefore collar-invariant. Transfer values are the two ordered directions of each pair; ranks are out of the twenty directions on the equalised frame. As-drawn transfer is read from Table B9, collar transfer from `aoi_frame_transfer.csv`.

	Manavgat–Muğla	Bejís–Montiferru
Schoener’s D , mean 1-D	0.826 (highest)	0.479 (lowest)
per-feature D	0.77 to 0.89	0.23 to 0.77
transfer, frames as drawn	0.470, 0.401	0.594, 0.548
transfer, 10 km collar	0.551, 0.510	0.669, 0.624
rank of 20 on the collar, from the bottom	5th, 2nd	15th, 11th
target cells inside the AoA	0.875, 0.531	—

Appendix C. Appendix C. Limitations

Section 5.8 states the four limitations that bind the conclusions; all eleven are here. The rest of this appendix, which specifies the protocol, is released with the supplementary appendices rather than printed here.

1040 *Appendix C.1. C.5 Limitations, in full*

Section 5.8 states the four that bind the conclusions; all eleven are here.

(i) **No meteorological covariates** enter the models, so we cannot say how local skill and portability behave for a mixed thermal-plus-weather predictor set.

1045 (ii) **The same-geography comparison covers one region only**, and even there year and seasonal phase are confounded. That confound cannot be resolved in this study area, and the reason is specific: the two events sit

42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine
1050 burned cells against this design’s gate minimum of thirty. The positive-block count for the 2022 arm is separately below the floor this design sets itself, at eleven against sixteen (Appendix A(m)). Its 331 burned cells also leave the thermal reversals unresolved at interval level, and the pair holds place fixed but not population. Those two arms are additionally the only transfer
1055 directions here computed by us rather than read from the pipeline author’s frozen export, with his unmodified code and the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1 (Giglio et al., 2018), whose omission and commission characteristics (Boschetti et al., 2019) bound every model evaluated here. No second
1060 burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072
1065 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tol-**
1070 **erance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs.** Both the successes and the failures of Section 4.6 should
1075 be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.** It is where the conditional diagnosis bites hardest and where feature removal recovers most. Three candidates have now been tested and none survives: its meteorology, which was not extreme; the quality screening
1080 of its coarse thermal input, which propagates widely but moves no signed association by more than $+0.0003$ (Appendix A(v), Appendix A(e)); and the evaluation frame, which explains its elevation figure but not its transfer behaviour (Section 4.4). With one fire season per region the remaining candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every sentence
1090 in this paper that leans on an exact count of supported directions should be read at that precision. (x) **The five areas of interest are not comparable frames, and this cohort cannot fully repair it** (Section 4.4). We report the equalised arm alongside the frame-as-drawn arm rather than replacing one with the other, because the collar radius is itself a choice and 5 km and 10
1095 km do not agree exactly (0.608 against 0.616). The deeper limitation is that the frames were fixed upstream of this work, in `repo/`, so we can restrict them but not extend them; a region whose rectangle is already fire-scale,

Montiferru, cannot be given a far field for symmetry. Any future cohort should fix the frame by an explicit accessible-area rule (Barve et al., 2011) before any predictor is computed, and we treat that as the main design lesson of this paper.

(xi) **One classifier family.** The headline numbers use a random forest with unlimited depth, the configuration most able to encode local structure and least able to extrapolate. Appendix A(h) shows the transfer result is not an artefact of that choice: three further estimators, including a penalised linear one, all land between 0.510 and 0.556 and all place fourteen of twenty directions above chance. Those are point estimates without intervals, so the ordering among them is not claimed as a result. Other model families were not tried, and a different inductive bias might behave differently, but within this family the negative result is a property of the predictors rather than of an unregularised estimator.

Declarations

Funding. This work was supported by the Çukurova University Scientific Research Projects Coordination Unit (Bilimsel Araştırma Projeleri Koordinasyon Birimi) under the Career Starter Project (Kariyer Başlangıç Projesi) scheme, project code FKB-2025-17608 (“Termal Dijital İkiz Tabanlı Sürü İHA Sistemi ile Orman Yangınlarının Erken Tespiti ve Önlenmesi”). Article processing charges are covered under the Çukurova University / TÜBİTAK EKVİL open-access agreement.

Acknowledgments. The authors gratefully acknowledge the Çukurova University Scientific Research Projects Coordination Unit for financial

support of this research, and the Department of Computer Engineering at Çukurova University for providing the laboratory environment and institutional support that made this work possible.

1125 **Competing interests.** The authors declare no competing interests.

Ethics approval. Not applicable. This study involved no human participants, animal subjects, or personally identifiable data.

Data availability. All satellite inputs are public. Burned-area labels are MODIS MCD64A1 Collection 6.1 and land cover is ESA WorldCover, 1130 both retrieved through Google Earth Engine; no proprietary or restricted data were used. The frozen numeric outputs behind every table and figure, together with the analysis code that reads them, are available at <https://github.com/dryuemco/thermal-twin>, which also carries the supplementary appendices: Appendix A and protocol sections C.1 1135 to C.4, C.6 and C.7, released with the paper rather than printed in it, at `paper/supplementary_appendices.md`. The upstream processing pipeline that produced those outputs is a separate public release, <https://github.com/emrehann17/satellite-thermal-digital-twin>, and the commit of record for every number reported here is 6381f4c. The 1140 modelling dataset each number rests on is identified there by SHA-256, which is how the one provenance incident in this project was settled: a rebuild had replaced one region’s file at its canonical path, and the recorded hash identifies the frozen original unambiguously. No digital object identifier is minted and no archival deposit exists.

1145 **Code availability.** The analysis code, its configuration and the scripts that regenerate each reported artefact are in the first repository above. Every

quantity in the paper can be regenerated from the released code and the frozen outputs. Three components were outside an earlier release and are now inside it: the reproduction-check driver and its validation package, on which
1150 Section 3.14 rests; the few-shot run, whose commit is reachable through the tag `few-shot-run-19d825b`; and the ERA5-Land diagnostic, whose manifest records commit `a07ea33`.

Author contributions. Stated in CRediT terms. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and
1155 editing. **Yunus Emre Cogurcu:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision.

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